

Xiaochuan Li

M.Sc. Human and AI · University of Technology Nuremberg

xiaochuan.li@utn.de · github.com/xchuan-li · Nürnberg, Germany

EDUCATION

M.Sc., Human and AI

2025 — 2027

University of Technology Nuremberg (UTN)

Research focus. Whether language models build the modal concepts semantic theory assigns — possibility and necessity as quantification over alternative worlds — or only their surface. Tested by dissociating the modal *word* from the modal *base* under controlled intervention, across typologically diverse languages.

Coursework. Cognitive psychology · interpretability · deep learning · epistemology.

German language program

2024 — 2025

University of Marburg · intensive German for graduate study

B.A., Philosophy

2020 — 2024

Shenzhen University · College of Humanities

Senior thesis: *On Leibniz's Logical Theory* (advisor: Zang Yong)

Coursework: logic, history of philosophy, analytic philosophy, mathematical logic; electives in cognitive psychology and linguistics. Shapes my focus on reasons, causality, and what licenses an inference.

PAPERS & PLANNED OUTPUTS

GRADUS

dataset + diagnostic paper · in progress

Does a language model represent epistemic modal force — the necessity in *must*, the possibility in *might* — or ride the surface modal word? A causal probe plus a graded epistemic-force dataset: a calibration diagnostic and a citable resource the modal-concepts thesis reuses.

[Project page](#) ↗

PROJECTS

HanGL — surface Hangul, latent Hanja

collaborative

The structure-versus-surface question on real LLMs via a natural phenomenon: Sino-Korean words, where Hangul shares no characters with its Hanja origin yet matches it in meaning. A wrong-Hanja intervention shows latent grounding scales with model capability — strong models neither need the cue nor are misled by it; a weak one is both. Pilot complete; scaling up.

[Project page](#) ↗ · [GitHub](#) ↗

Latent Control States

collaborative

Does prompt framing — urgency, authority, epistemic positioning — only change an LLM's outputs, or shift a causally relevant latent state? The controlled two-agent data pipeline is built; linear probing and activation patching come next. Advised by a senior researcher at DFKI (the German Research Center for Artificial Intelligence).

[Project page ↗](#)

Cross-lingual ProtoBias

collaborative

An applied multimodal stress test: does a vision-language model track semantic content, or language- and culture-specific prototype shortcuts? Extends a forthcoming paper from an established NLP group with a cross-lingual evaluation, now in its second iteration.

[Project page ↗](#) · [GitHub ↗](#)

CAST

personal · workbench

A workbench that stipulates a formal-semantic object and projects it to text — grammatical structure, the inference it licenses, and interventions made observable — so structure recovery is tested against a known target. First domain: the causal graph; being generalised into a modal-semantic compiler for the thesis.

[Project page ↗](#)

SKILLS

Both the engineering my research needs — controlled linguistic data generation, transformer model evaluation, intervention-based causal testing, reproducible pipelines — and the foundations it draws on: cognitive psychology, linguistics, and logic.

Programming & data

Python (NumPy, pandas, scikit-learn), Git, LaTeX.

ML / NLP tooling

PyTorch and HuggingFace Transformers (working proficiency) — model inference, dataset pipelines, LoRA fine-tuning.

Causal & interpretability methods

Controlled do-interventions, causal graphs, and shortcut-severing audits; developing behavioral path-tracing.

Cognitive psychology

Working foundations in concept learning, cognitive development, and human reasoning — including the developmental literature on modal cognition (Leahy & Carey), which grounds the emergence question the thesis asks.

Linguistics & logic

Formal semantics (possible-worlds, modal quantification), modal and predicate logic, and morphology/syntax across isolating, fusional, and agglutinative types; grounded in a philosophy background in logic and philosophy of language.

Linguistic evaluation

Designing controlled stimuli for morphology, agreement, dependency structure, and word order across Chinese, German, English, and Japanese.

LANGUAGES

Mandarin Chinese (native) · German (C1) · English (C1) — spanning isolating, fusional, and analytic types. Learning Japanese, an agglutinative head-final system, to extend the typological span my linguistic work draws on.